What Li Has Been Up To

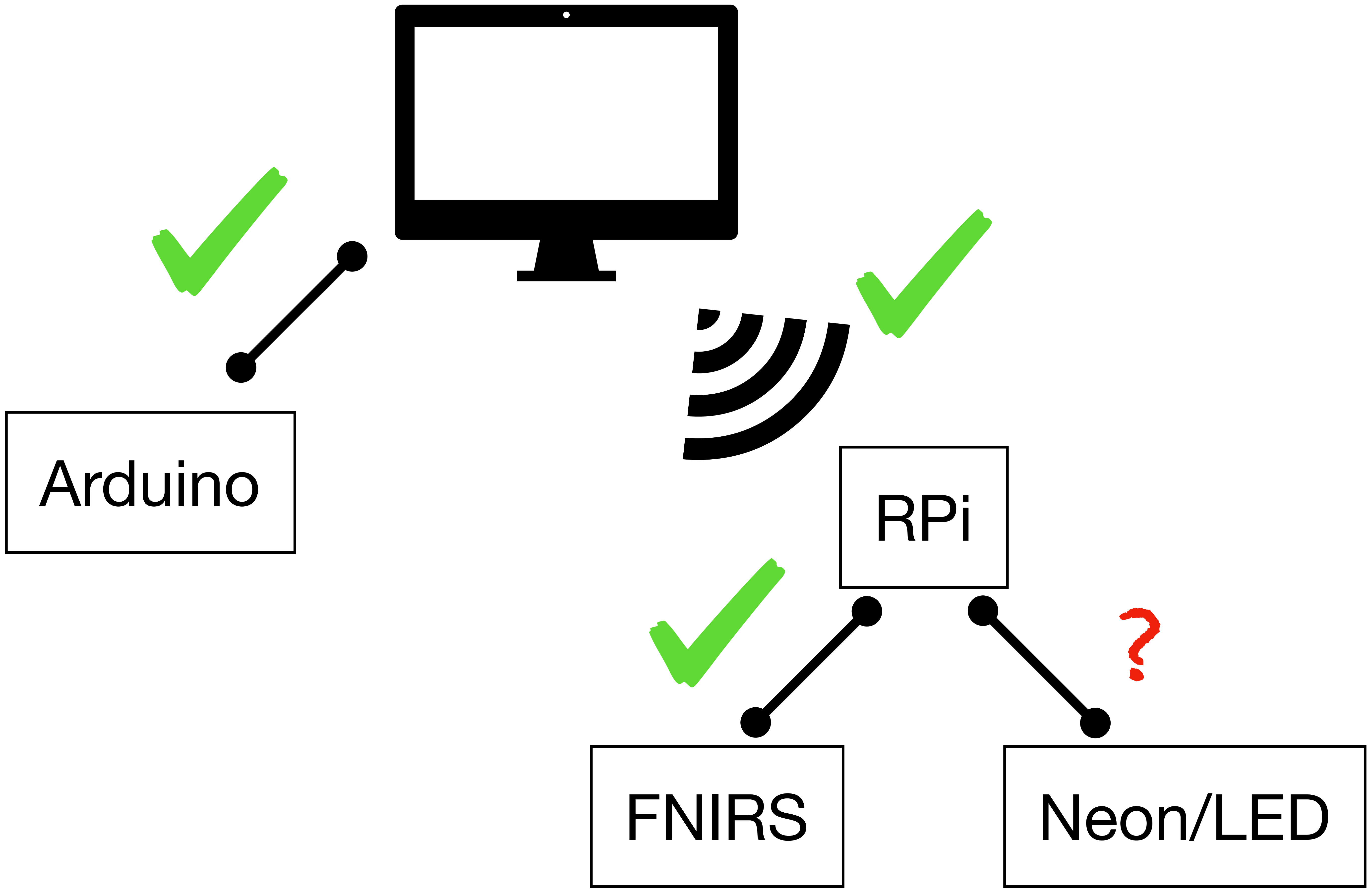
Friday, July 18, 2025

Overview

1. Eye Tracking LED alignment verification
2. LSL (**L**ab **S**treaming **L**ayer) integration and verification
3. Boas Lab IMUs
4. Documentation for all of the above

1. Verifying Eye Tracking Alignment

- Already verified by Sung:
  - Unity to RPi
  - Unity to Arduino
  - RPi to FNIRS
- To Do: RPi to Eye tracking with LED solution
  - *should* be okay since RPi to FNIRS was ok



Results

Run Unity project

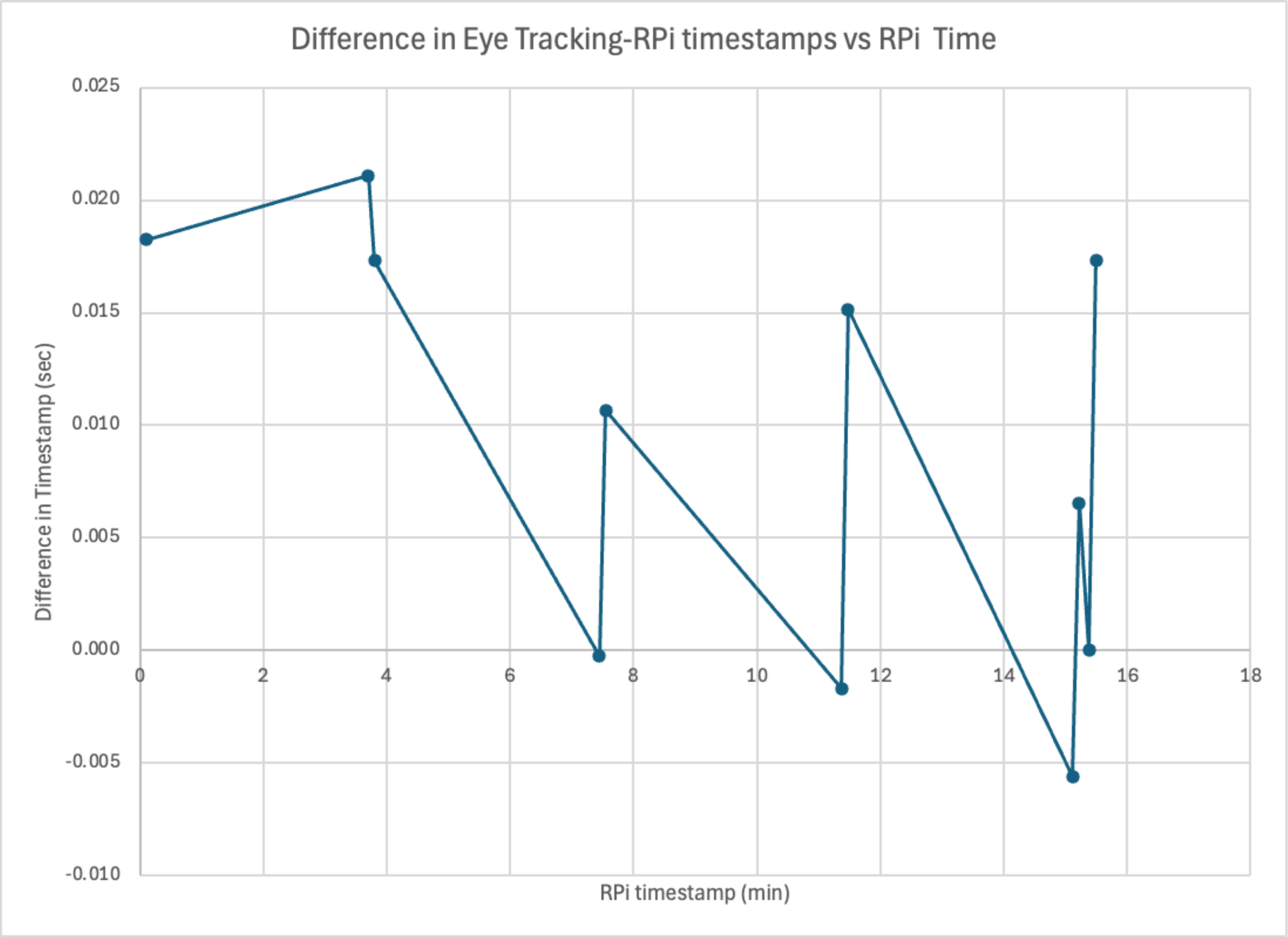
Find timestamps for when the LED turns on

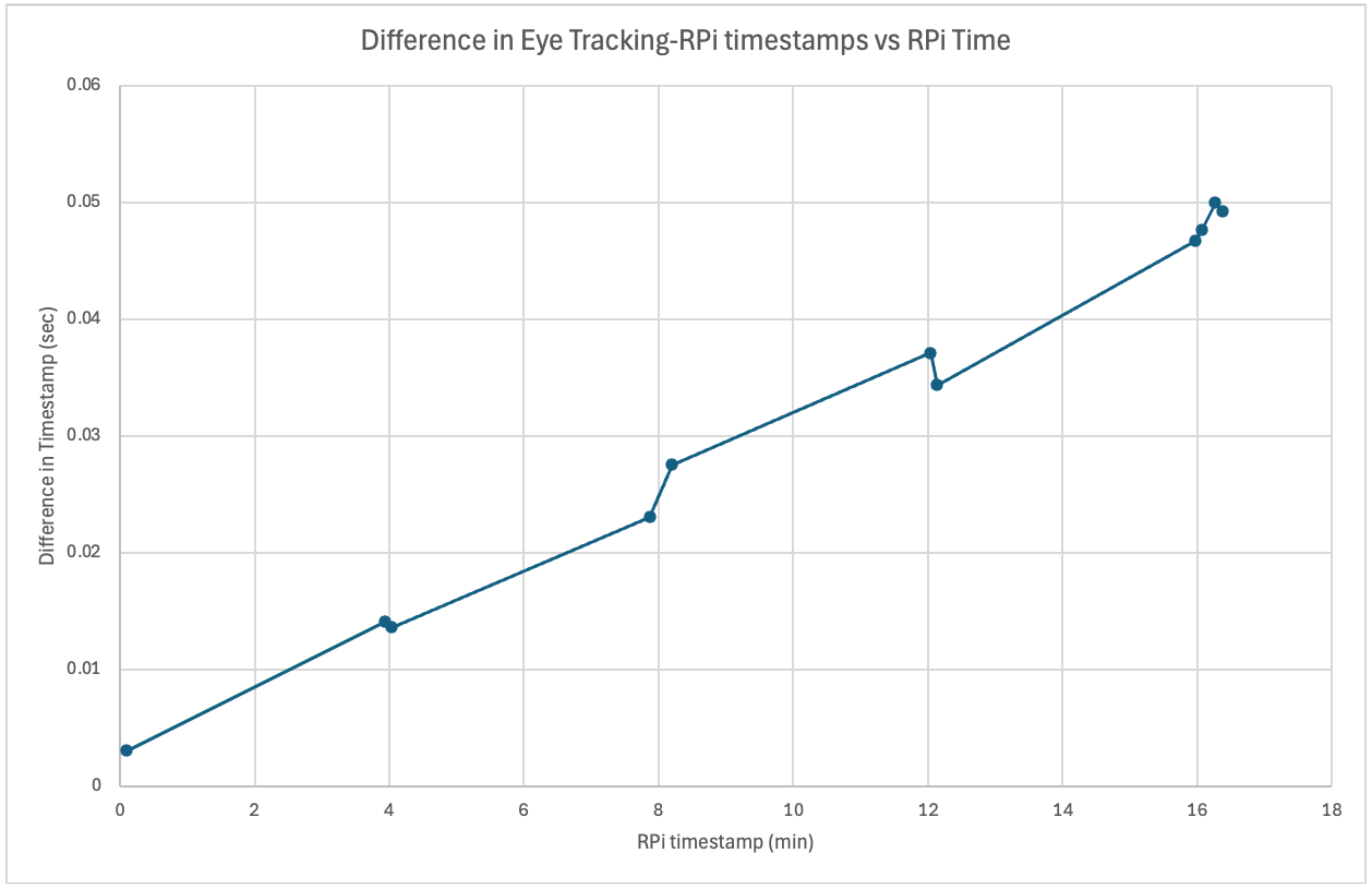
Compare differences in timestamps from RPi vs eye tracking

Frame rate:

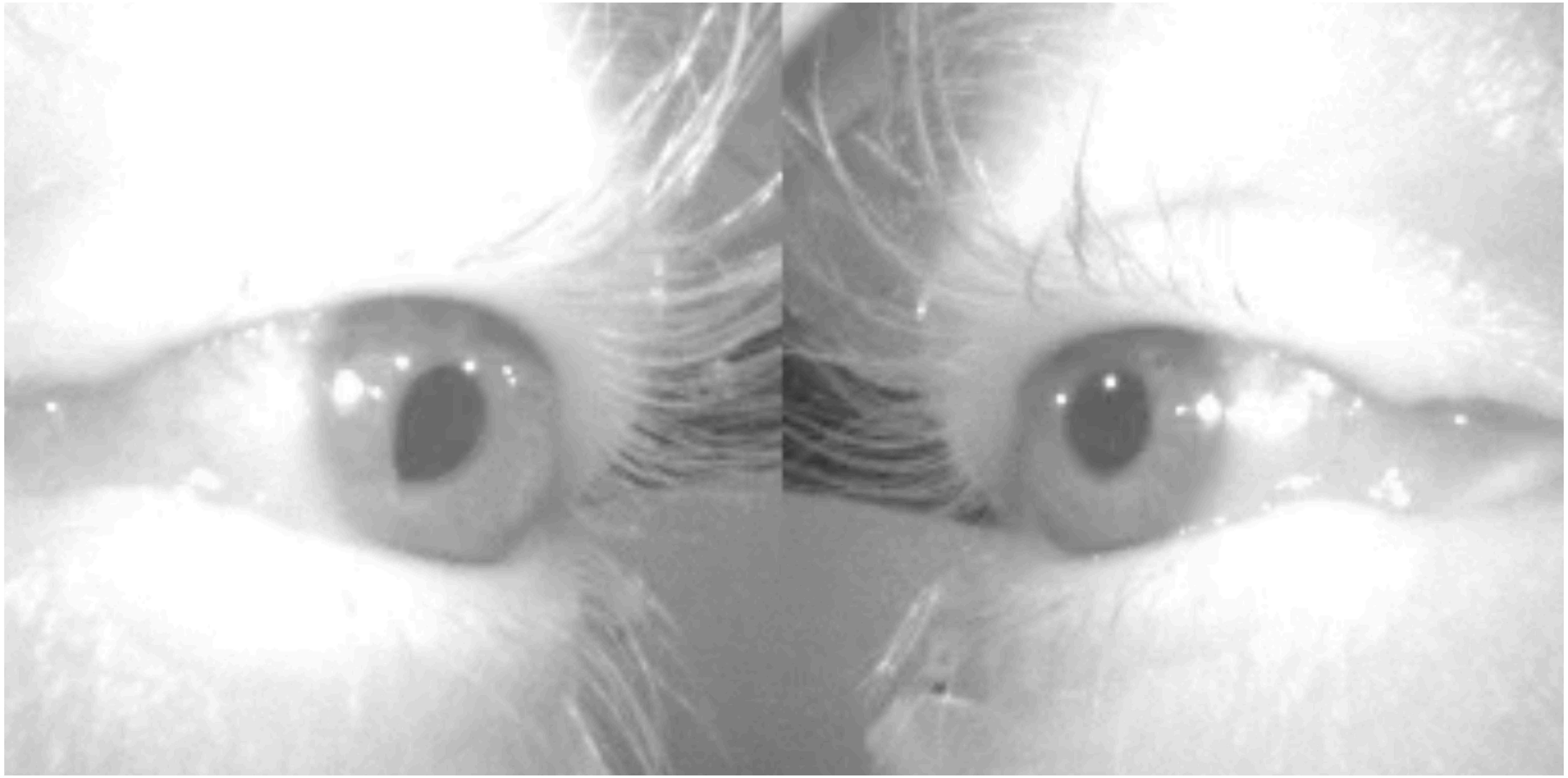
Environment camera: 30 Hz

Eye cameras: 200 Hz





Problem with LED solution



2. Lab Streaming Layer (LSL)

(Not a pro, so I don't know the nitty gritty yet)

1. What is it?

A system for unified collection of measurement time series

It handles:

- Networking
- Time synchronization
- Data storage of recorded streams

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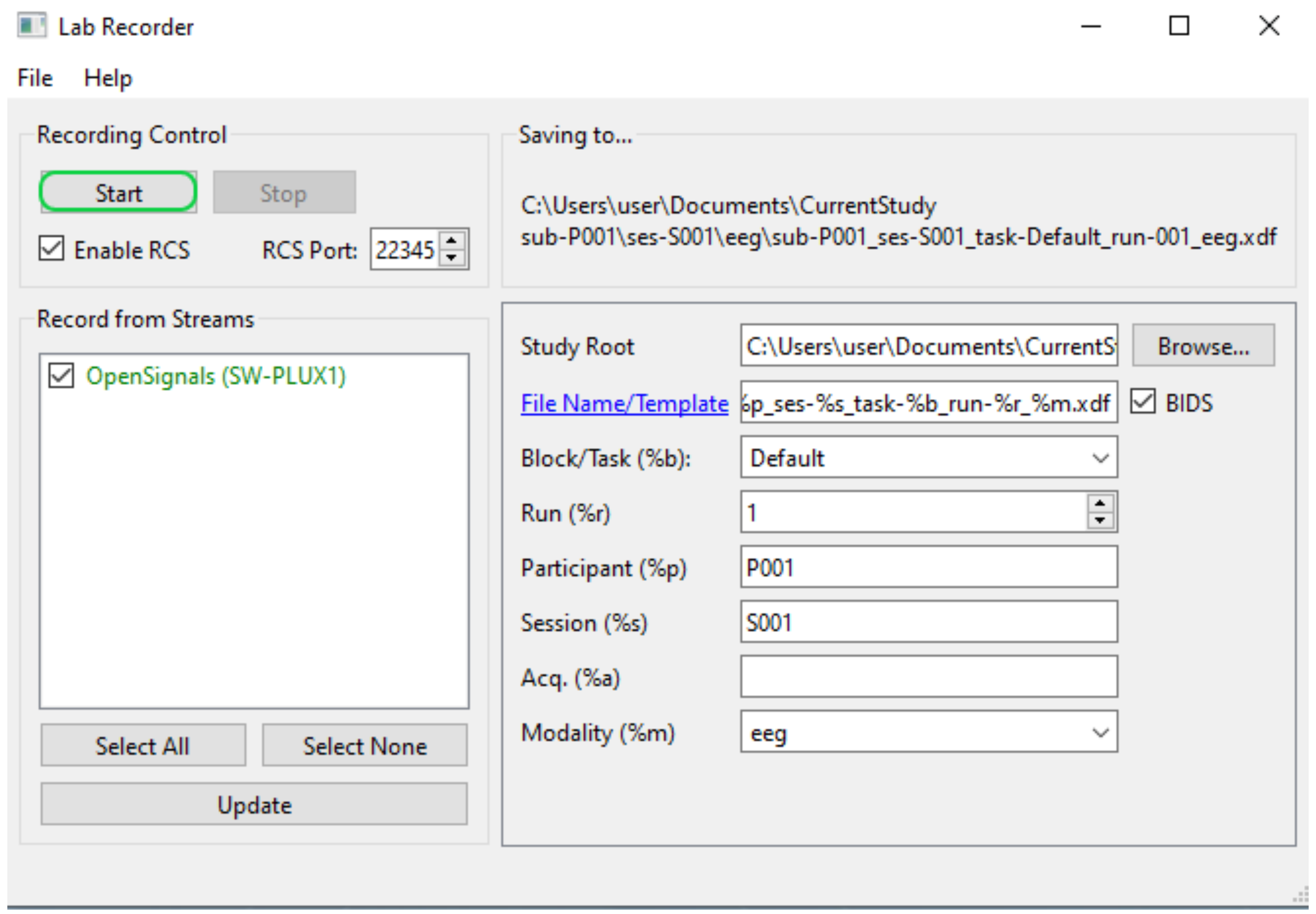
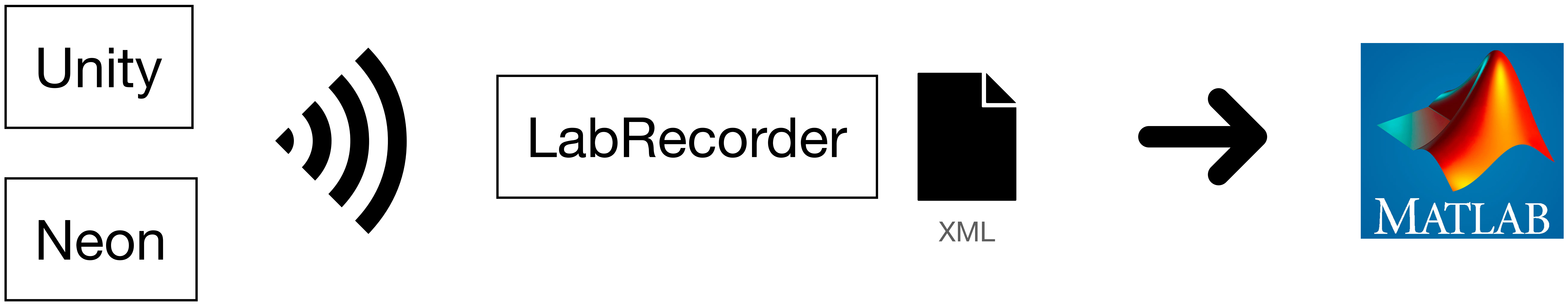
2. How will it be used in our system?



https://support.pluxbiosignals.com/knowledge-base/how-do-i-stream-data-through-the-labrecorder-using-opensignals-lsl-feature-and-python/

LSL Outlets

LSL Inlet



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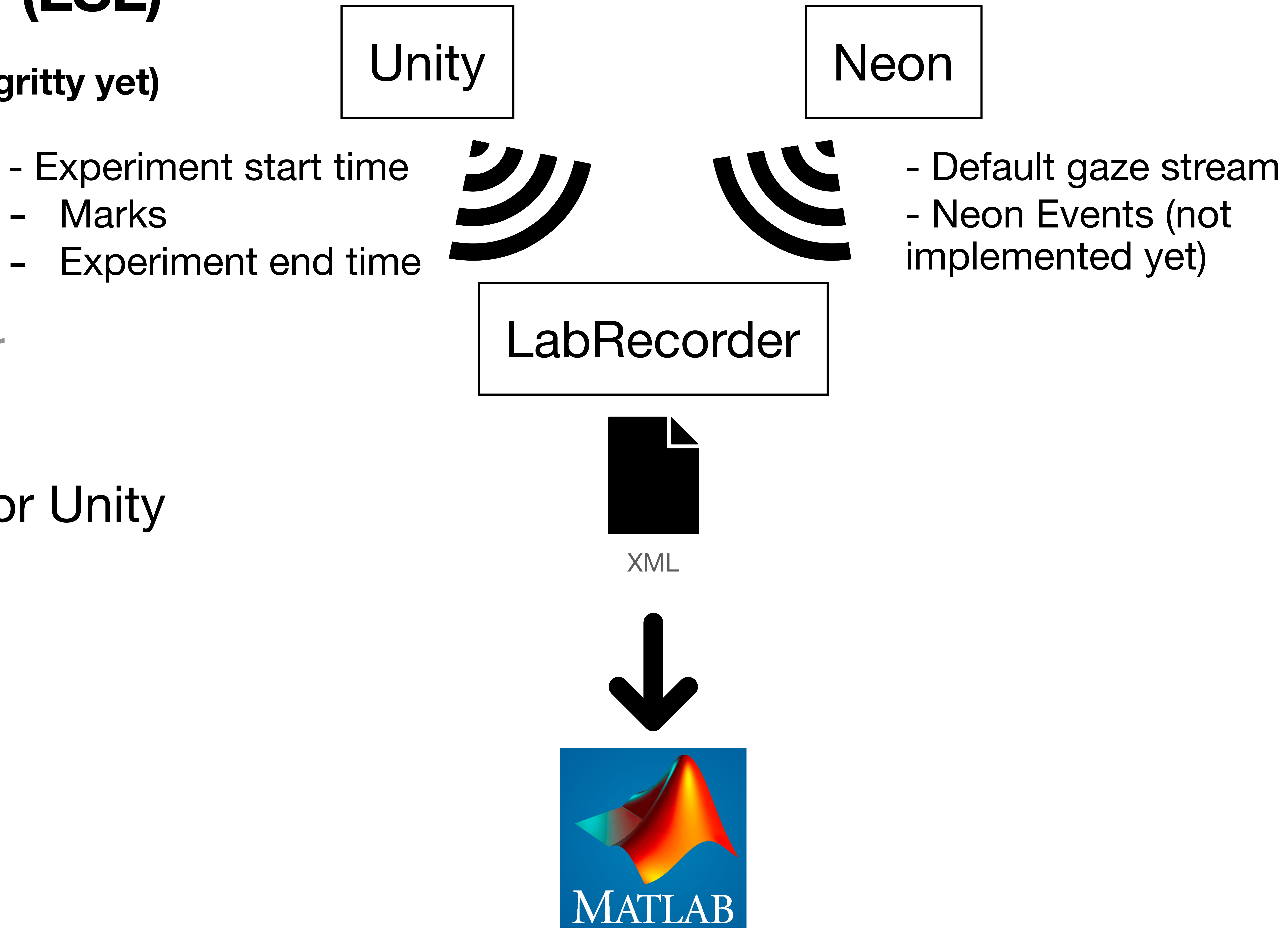
2. How will it be used in our system?
3. Progress: implemented for Unity and Neon



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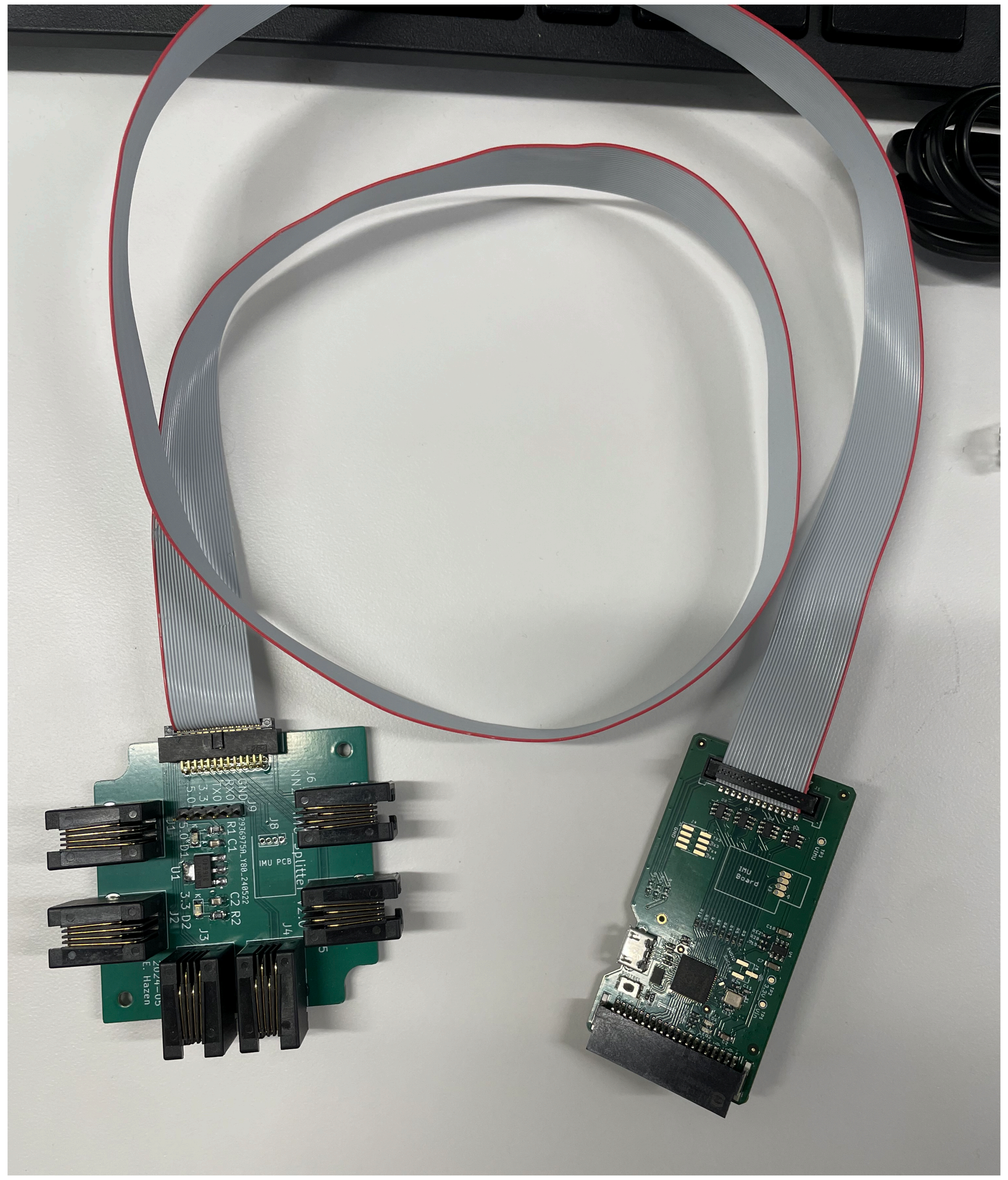
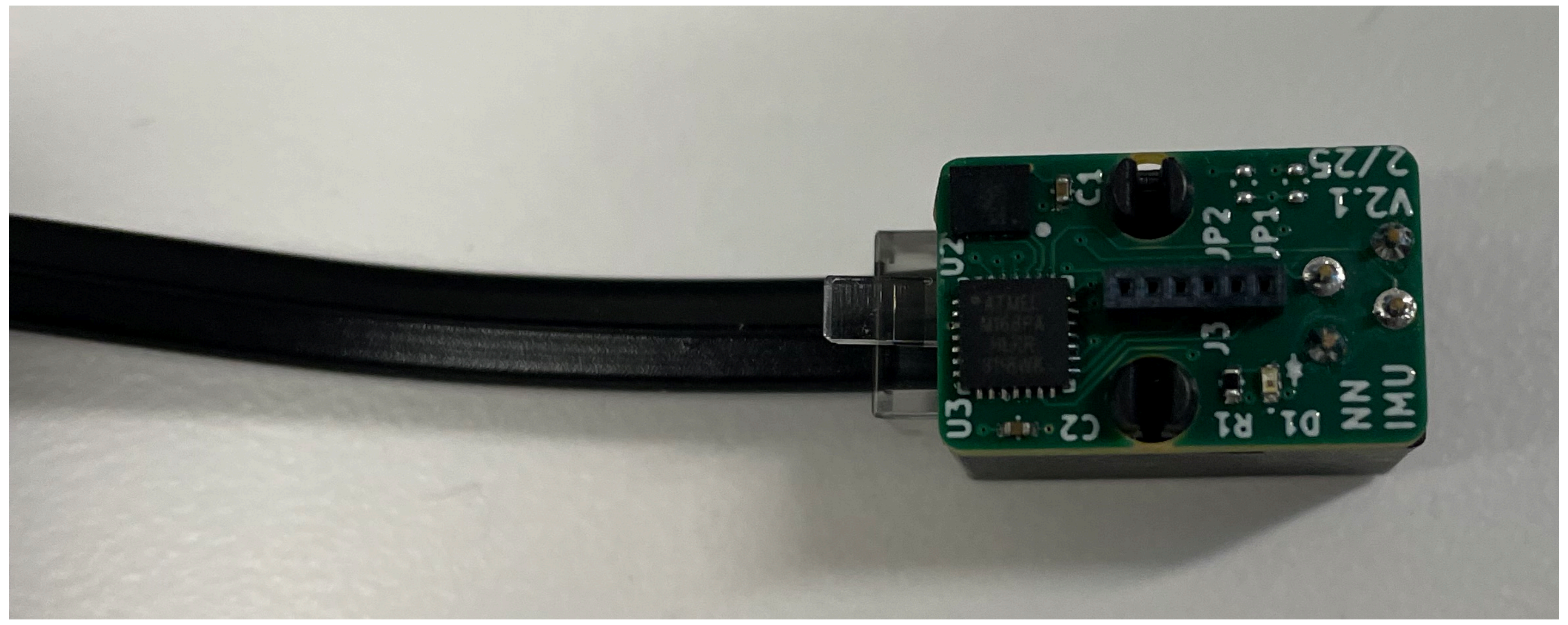
1. What is it?
2. How will it be used in our system?
3. Progress: implemented for Unity and Neon
4. Next steps

1. Verify accuracy
    - RPi LSL mark when Unity sends to RPi and after actual mark occurs
    - Implement LSL for RPi (python)
    - Run experiment with LSL and LED solutions
2. Add Neon Events
    - Neon only records limited data via LSL, other data must be captures with the API and manually aligned using Neon Events ("mark" equivalent")
3. Other stuff will probably pop up

3. Boas Lab IMUs

- Data output rate is 20 Hz
- Need cases for all components
- Need to test sending a mark to the IMUs
- Data analysis/validation
- Plan to set these IMUs up then set up Movella dot to compare systems





Output data:

Channel (0-5), temp, gyroscope x,y,z, acceleration x,y,z timestamp, signal T/F

4. Documentation

Just writing up everything I’m learning and how to set things up from scratch